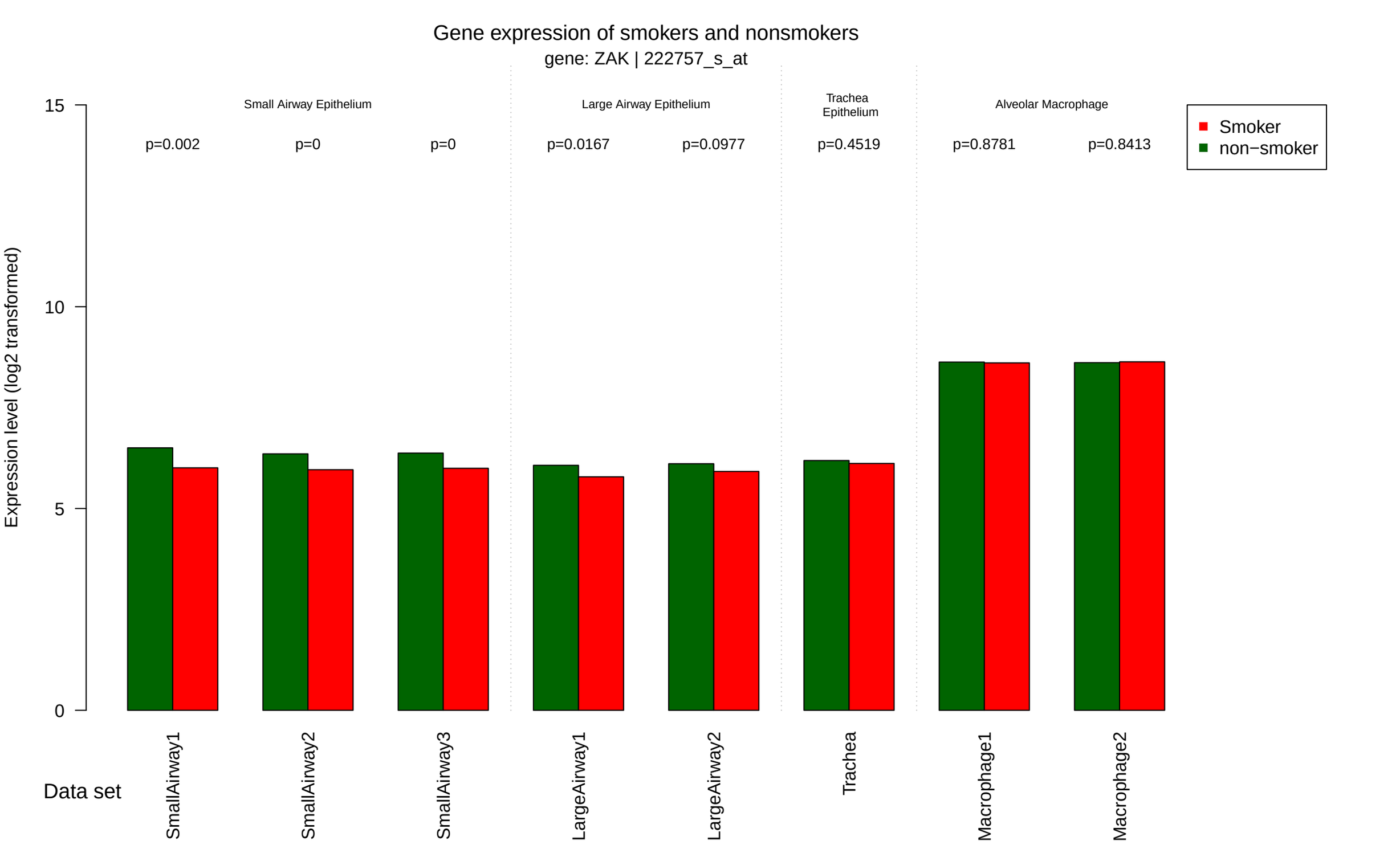
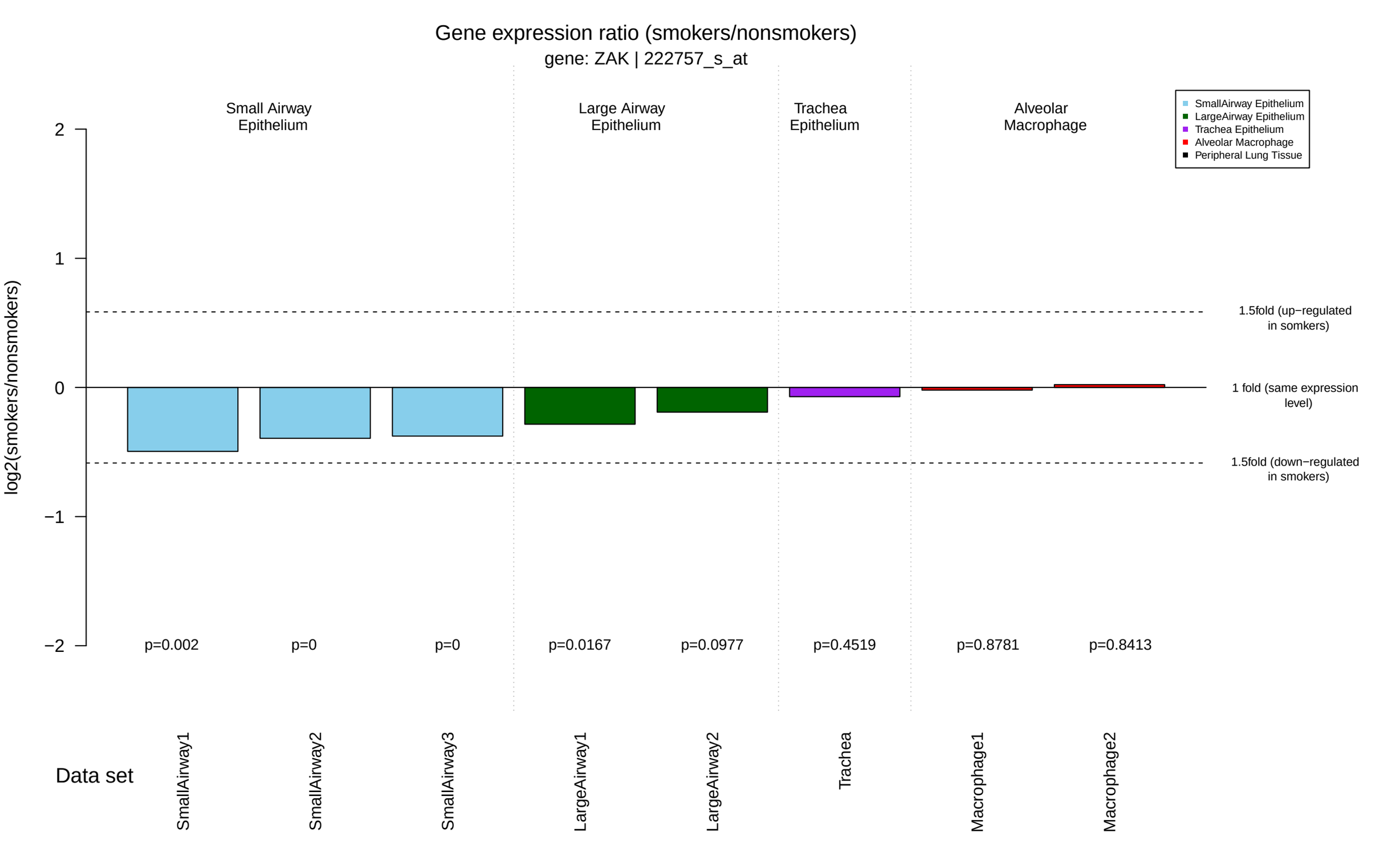
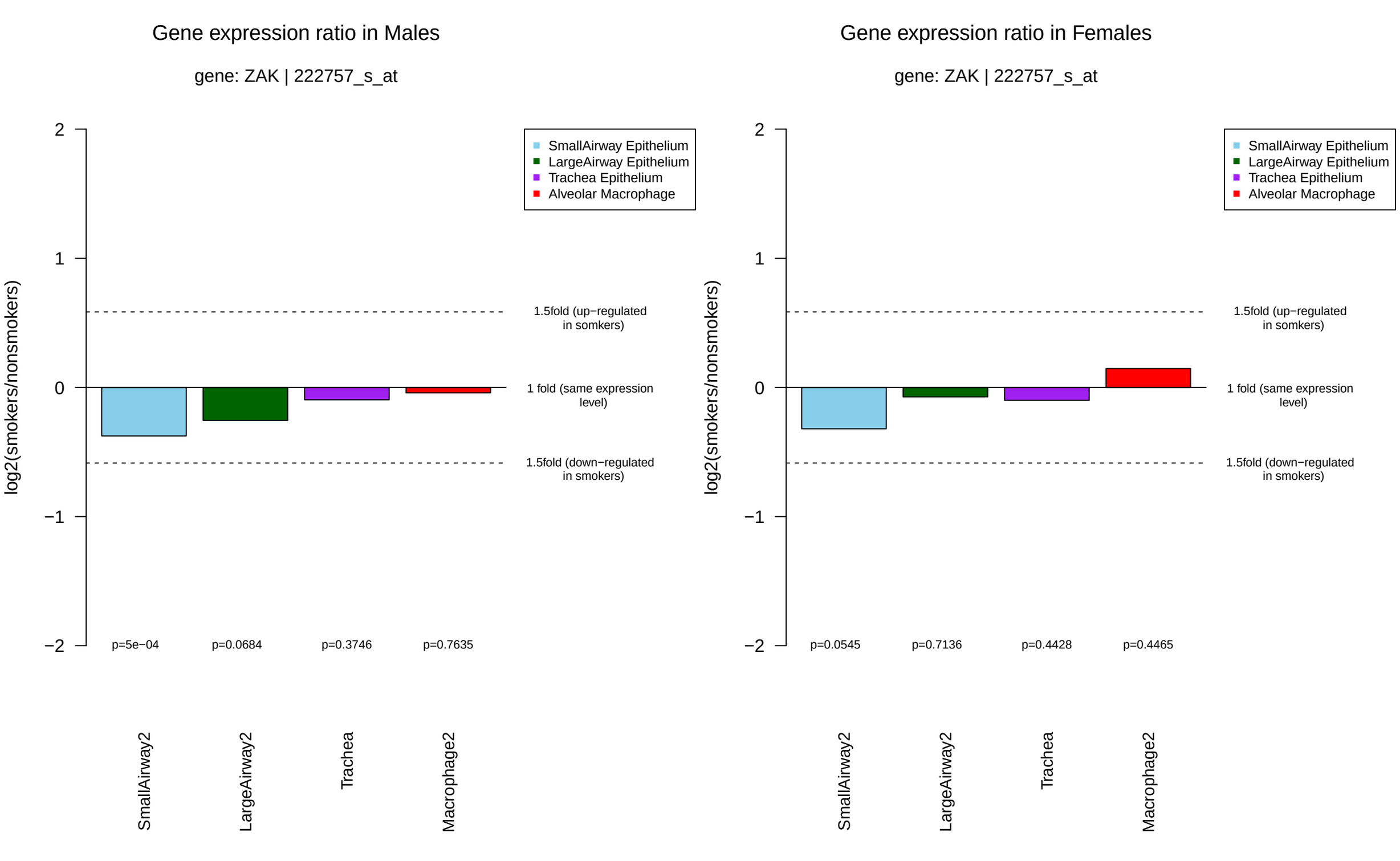
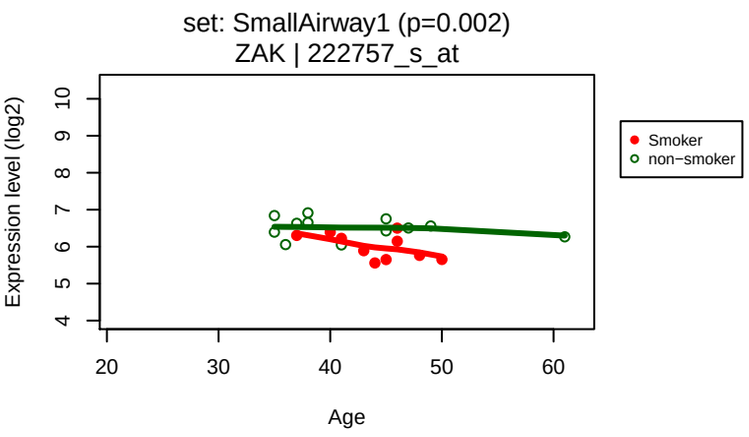
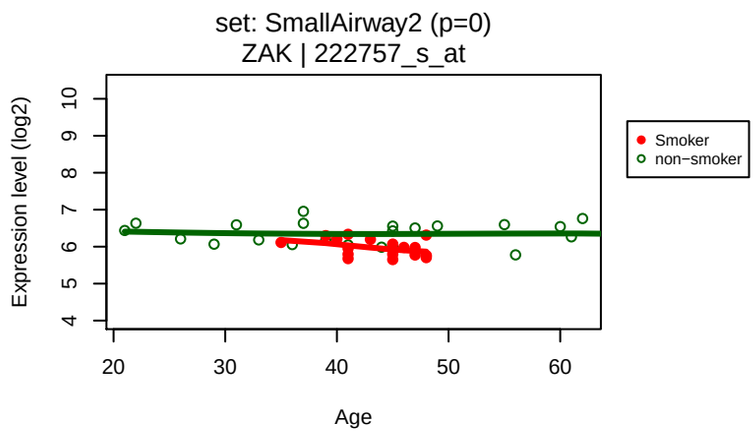
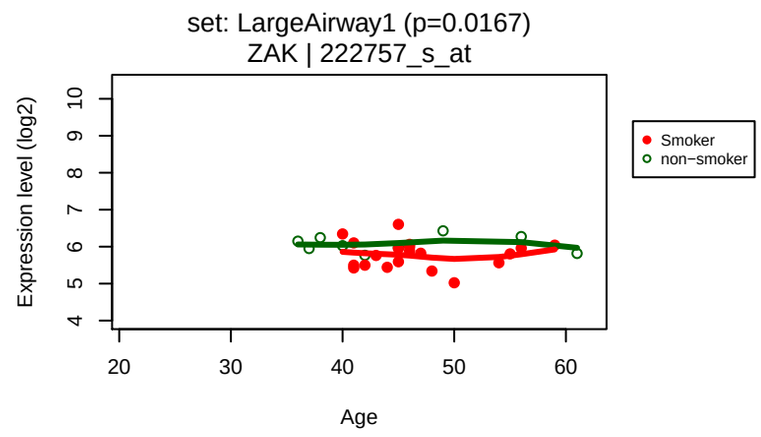

Gene expression of smokers and nonsmokers
gene: ZAK | 222757_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.002
p=0
p=0
p=0.0167
p=0.0977
p=0.4519
p=0.8781
p=0.8413
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

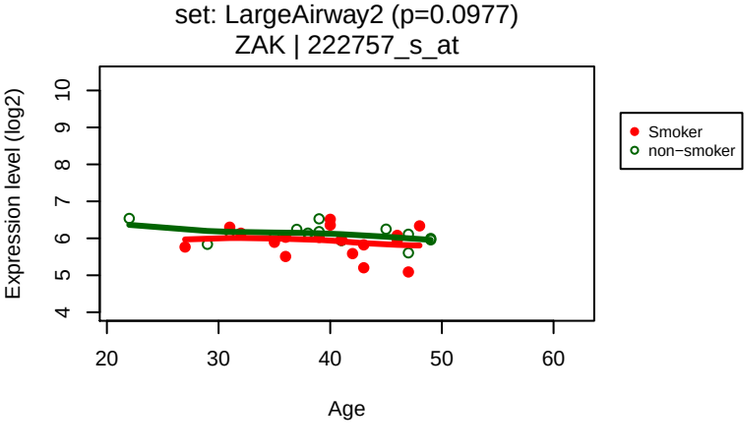

Gene expression ratio (smokers/nonsmokers)
gene: ZAK | 222757_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.002
p=0
p=0
p=0.0167
p=0.0977
p=0.4519
p=0.8781
p=0.8413
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

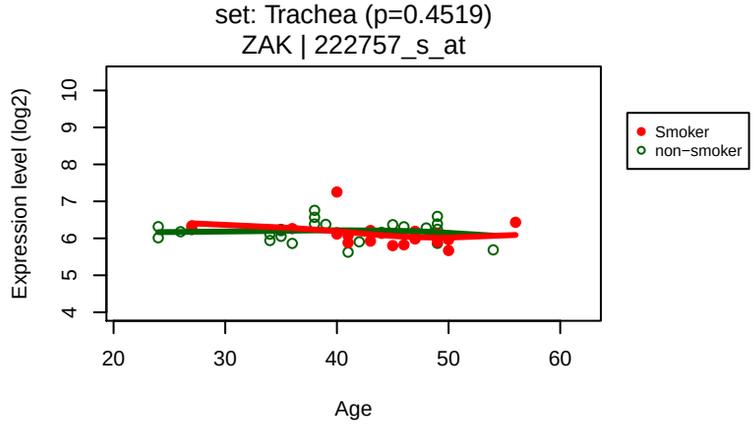

Gene expression ratio in Males
gene: ZAK | 222757_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=5e−04
p=0.0684
p=0.3746
p=0.7635
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ZAK | 222757_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0545
p=0.7136
p=0.4428
p=0.4465
SmallAirway2
LargeAirway2
Trachea
Macrophage2

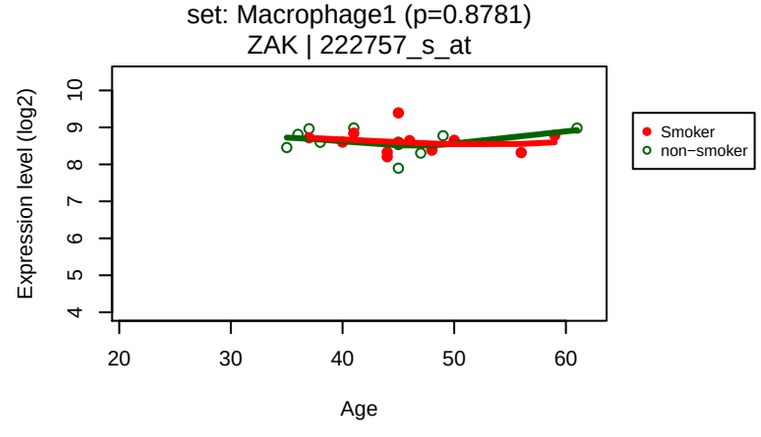

set: SmallAirway1 (p=0.002)
ZAK | 222757_s_at
Expression level (log2)
Age
Smoker
non−smoker

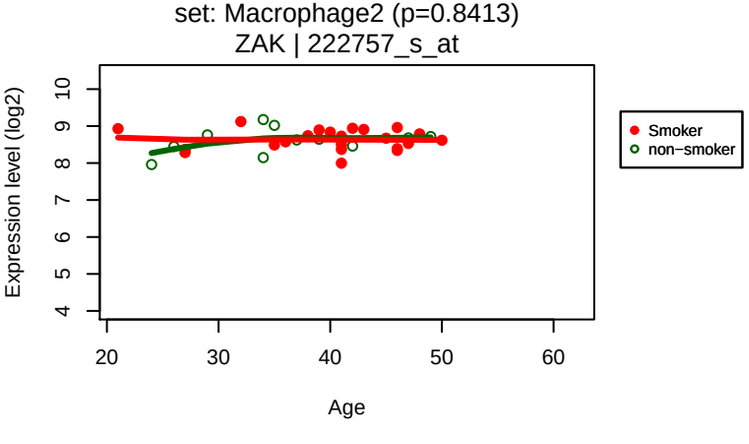

set: SmallAirway2 (p=0)
ZAK | 222757_s_at
Expression level (log2)
Age
Smoker
non−smoker

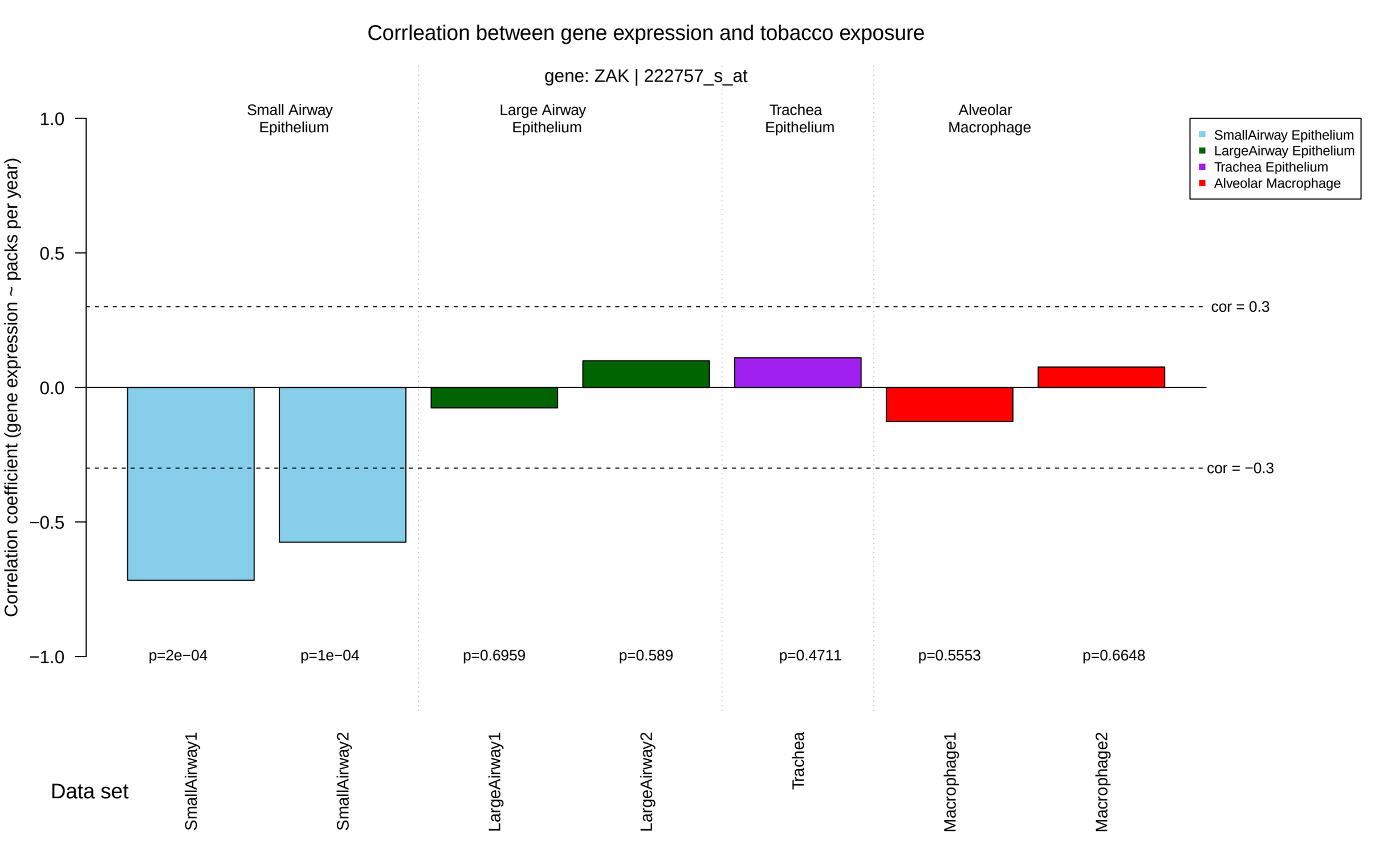

set: LargeAirway1 (p=0.0167)
ZAK | 222757_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0977)
ZAK | 222757_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4519)
ZAK | 222757_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8781)
ZAK | 222757_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8413)
ZAK | 222757_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ZAK | 222757_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=2e−04
p=1e−04
p=0.6959
p=0.589
p=0.4711
p=0.5553
p=0.6648
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2